

Objective Benchmarking of Vietnamese Zero-Shot Voice Cloning: F5-TTS ViVoice Against Five Text-to-Speech Baselines

Group 4 · Data Science Capstone Project (DSP391m)

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ABSTRACT

Voice cloning text-to-speech (TTS) must satisfy two competing objectives at once: the synthesized speech must be *intelligible* (it says the right words) and must *preserve speaker identity* (it sounds like the reference voice). We present a reproducible, objective benchmark of six TTS systems on Vietnamese, centered on **F5-TTS ViVoice**, a Diffusion-Transformer model with a Conditional Flow-Matching objective and a Vocos 24 kHz vocoder, evaluated against XTTS-v2, MMS-TTS, Piper, Edge-TTS, and Bark.

Because generative speech has no single ground-truth label, we quantify quality with automatic proxies: Word/Character Error Rate (WER/CER) obtained by an automatic-speech-recognition (ASR) round-trip through Whisper, speaker similarity (SECS) via cosine distance over Resemblyzer embeddings, and generation speed via the real-time factor (RTF). Under a fixed sentence, seed, and reference sample, F5-TTS attains the lowest WER (**5.9%**) while remaining in the top tier of speaker similarity (**0.837**) — the only engine that jointly optimizes both axes. We report the full trade-off, discuss where each engine belongs, and state the study's limitations candidly, including a pilot-scale test set and the fact that the fine-tuning pipeline is hardened but not yet verified end-to-end on real speaker data.

Index Terms — Vietnamese TTS, zero-shot voice cloning, flow matching, diffusion transformer, objective evaluation, WER, speaker similarity.

1 Introduction

Zero-shot voice cloning takes a short reference recording of an unseen speaker and a target text, and reads that text in the reference speaker's voice — without any per-speaker training. It underpins accessibility tools, dubbing, and personalized assistants, but Vietnamese remains under-served relative to English: the

language's six lexical tones and dense diacritics make both synthesis and its evaluation harder, and a mispronounced tone can change word meaning entirely.

This report addresses the *Model & Results* stage of the capstone. Our contributions are: (i) a system built around F5-TTS ViVoice with a corrected, library-faithful inference pipeline and a production-hardened fine-tuning pipeline; (ii) an **objective, reproducible evaluation harness** that turns the vague notion of "sounds good" into measurable quantities (WER, CER, SECS, RTF, and audio-signal metrics) computable fully offline; and (iii) a head-to-head benchmark of six engines under identical conditions, with an explicit analysis of the intelligibility–identity–speed trade-off and an honest account of what the numbers can and cannot support.

2 Background and Related Work

Modern non-autoregressive TTS has converged on transformer backbones paired with generative objectives. **F5-TTS** [1] uses a Diffusion Transformer (DiT) [3] trained with a **Conditional Flow-Matching** (CFM) objective [2], which learns a continuous vector field transporting noise to the target mel-spectrogram distribution; a **Vocos** neural vocoder [4] then converts mel-spectrograms to a 24 kHz waveform. The Vietnamese checkpoint *hynt/F5-TTS-Vietnamese-ViVoice* [10] is pre-trained on roughly one thousand hours of Vietnamese speech.

For comparison we include **XTTS-v2** [5], a competing multilingual zero-shot cloning model; **MMS-TTS** [6], Meta's single-speaker Vietnamese VITS; **Piper** [9], a fast local ONNX voice; **Edge-TTS**, Microsoft's cloud neural voice used as a high-quality single-speaker baseline; and **Bark** [8], included as a deliberately weak lower bound because it has no Vietnamese support. Evaluation relies on **Whisper** [7] for the ASR round-trip and on a GE2E-style speaker encoder [11] (Resemblyzer) for identity similarity.

3 System and Model Architecture

3.1 Model selection

F5-TTS ViVoice was selected as the primary model because it uniquely combines zero-shot cloning (a short reference suffices, with no per-speaker fine-tuning), strong Vietnamese pre-training, a state-of-the-art DiT+CFM backbone that avoids the monotone quality of single-speaker VITS, and a research-permissive license (CC-BY-NC-SA-4.0) suitable for a non-commercial academic project. The five remaining engines were chosen to span the trade-off space, from competing cloning (XTTS-v2) to fast fixed-voice TTS (MMS, Piper, Edge) to a weak floor (Bark).

3.2 Architecture

The architecture is loaded verbatim from the library's `F5TTS_Base.yaml` so that it matches the pre-trained checkpoint exactly:

```
CFM (Conditional Flow-Matching) wrapper
└─ DiT backbone: dim=1024, depth=22, heads=16, ff_mult=2,
```

```
text_dim=512, conv_layers=4,  
text_mask_padding=false, pe_attn_head=1 ← compatibility-critical  
└ Vocoder: Vocos @ 24 kHz (mel → waveform)  
Parameters ≈ 336M · Audio front-end: n_fft=1024, hop=256, win=1024, n_mel=100
```

COMPATIBILITY INVARIANT

The two fields `text_mask_padding=false` and `pe_attn_head=1` must not be altered. When omitted, DiT falls back to v1 defaults and the forward pass silently mis-computes despite correctly-shaped weights, degrading output to a near-monotone signal (spectral flatness collapses to ≈ 0.0019). Restoring the exact architecture returns spectral flatness to the human-speech range (≈ 0.11 – 0.14). This was one of the two principal defects diagnosed and fixed during development.

3.3 Inference pipeline

The F5-TTS path uses the library's official `preprocess_ref_audio_text` and `infer_process` utilities so that tokenization (`convert_char_to_pinyin`) and preprocessing match training. Long inputs are normalized (Unicode NFC, diacritics preserved), split at sentence/clause boundaries with URLs, e-mails, decimals, dates, and abbreviations protected from splitting, synthesized chunk-by-chunk under a fixed seed, joined with 180 ms of silence, and peak-normalized (≤ 0.99) to prevent clipping. Each engine lazy-loads and caches independently; an `RLock` plus a concurrency-1 queue prevents out-of-memory from concurrent model loads, and the system falls back to CPU if CUDA/MPS fails.

4 Data and Training Methodology

The fine-tuning objective is single-speaker adaptation while avoiding catastrophic forgetting of the pre-trained model. The pipeline (`scripts/train.py`) proceeds from a metadata manifest through a deterministic train/validation split, a padding-aware collate function, the CFM flow-matching loss, mixed-precision back-propagation with gradient accumulation, gradient clipping, and a manual warm-up-then-cosine learning-rate schedule; the best checkpoint is selected by validation loss.

Table 1. Training hyper-parameters (tuned for an NVIDIA T4, 16 GB).

HYPER-PARAMETER	VALUE	RATIONALE
Learning rate	1e-5	Small LR for fine-tuning; avoids forgetting
Batch × grad-accum	1 × 8 = 8	Effective batch under limited VRAM
Warm-up steps	200	Linear 0→LR for early stability
Scheduler	cosine	Anneal to min-LR 1e-7
Max grad norm	1.0	Gradient clipping
Mixed precision	fp16	T4 supports FP16; auto-off on CPU
Optimizer / seed	AdamW / 42	$\beta=(0.9,0.999)$, wd=0.01; deterministic

The pipeline is hardened for reproducibility and provenance: exponential-moving-average weights are loaded correctly from the official checkpoint; the train/validation split is deterministic under seed 42; NaN/Inf losses abort immediately; resumption is refused if metadata is missing or the base model mismatches; each checkpoint stores a JSON record of base model, architecture, vocabulary size, config, *git commit*, step, validation loss, and dataset summary.

HONEST STATUS

The fine-tuning pipeline is hardened and unit-tested at the component level, but **not yet verified end-to-end on real human-speaker data with a GPU**; the blocker is the absence of 30–60 minutes of clean, hand-transcribed Vietnamese speech. Earlier checkpoints trained on synthetic tones were quarantined and are *not* used to claim quality. Accordingly, the results in §6 characterize the **zero-shot base model**, not a fine-tuned one.

5 Evaluation Methodology

5.1 Metrics

Generative speech lacks a simple correctness label, so quality is measured through objective proxies on two core axes plus speed. **Intelligibility** is measured by WER and CER via an ASR round-trip: Whisper (`small` , beam 5, `language=vi`) re-transcribes the generated audio, and the transcript is compared to the reference text with edit distance (via `jiwer` , with an equivalent local fallback). Vietnamese diacritics are preserved by default. **Speaker identity** is measured by SECS: the cosine similarity between Resemblyzer embeddings of the generated and reference audio. **Speed** is the real-time factor, $\text{RTF} = \text{inference time} \div \text{audio duration}$. Auxiliary signal metrics (RMS, peak, clipping ratio, spectral flatness) screen for degenerate output.

INTERPRETATION GUARDRAILS

SECS is a *relative* similarity score, not biometric proof of identity. WER/CER are upper bounds on true error because Whisper itself errs, especially on tones — so the numbers are most meaningful *relatively*, when comparing engines under identical inputs, rather than as absolute rates.

5.2 Experimental setup

All six engines synthesize the *same* sentence — "Trí tuệ nhân tạo đang thay đổi cách con người giao tiếp và làm việc mỗi ngày." (17 words) — from the *same* reference sample under a fixed seed (42). The harness is fail-soft: each case carries its own status, a corrupt file or missing dependency does not abort the batch, and exit codes distinguish full success, total failure, and an invalid manifest. Cross-validation in the classical k-fold sense is impractical for fine-tuning a 336M-parameter generative model on tens of minutes of audio; its role is filled by a deterministic held-out validation loss during training and an independent fixed test set of diverse sentences (short, numeric, decimal, date, abbreviation, proper-noun, interrogative, compound, and long) for post-hoc metric evaluation.

6 Results

Table 2. Objective results, six engines, identical sentence / seed / reference. Bold marks the best value per column. Source: `outputs/evaluation/results.json` (unedited).

ENGINE	TYPE	WER ↓	CER ↓	SECS ↑	TIME (S)	RTF ↓
F5-TTS	voice cloning	0.059	0.026	0.837	193.0	45.12
XTTS-v2	voice cloning	0.118	0.065	0.888	36.2	8.48
MMS-TTS	fixed voice	0.118	0.039	0.601	3.9	0.61
Piper	fixed voice	0.118	0.052	0.615	2.1	0.59
Edge-TTS	cloud, fixed	0.118	0.065	0.561	3.0	0.65
Bark	fixed (no VI)	1.176	0.844	0.480	167.2	11.59

Intelligibility via ASR round-trip (Whisper small)

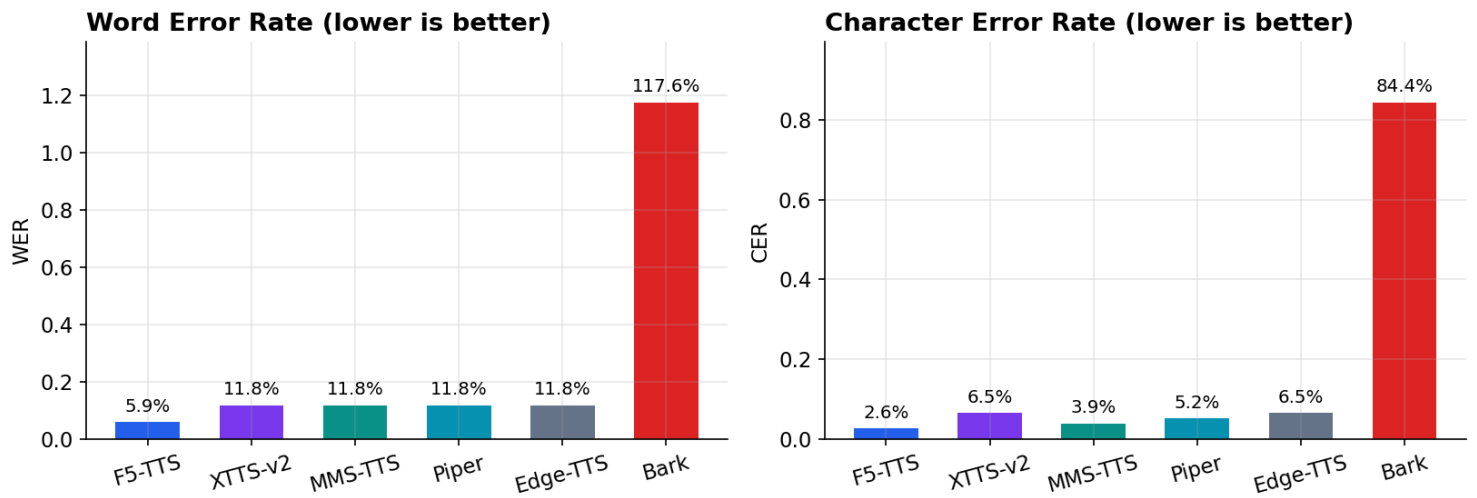


Figure 1. Intelligibility via ASR round-trip. F5-TTS attains the lowest WER (5.9%); the XTTS/MMS/Piper/Edge cluster sits near 11.8% (a two-word difference); Bark exceeds 100% because it cannot read Vietnamese, forming the lower bound.

Speaker similarity (SECS): generated vs. reference (higher is better)

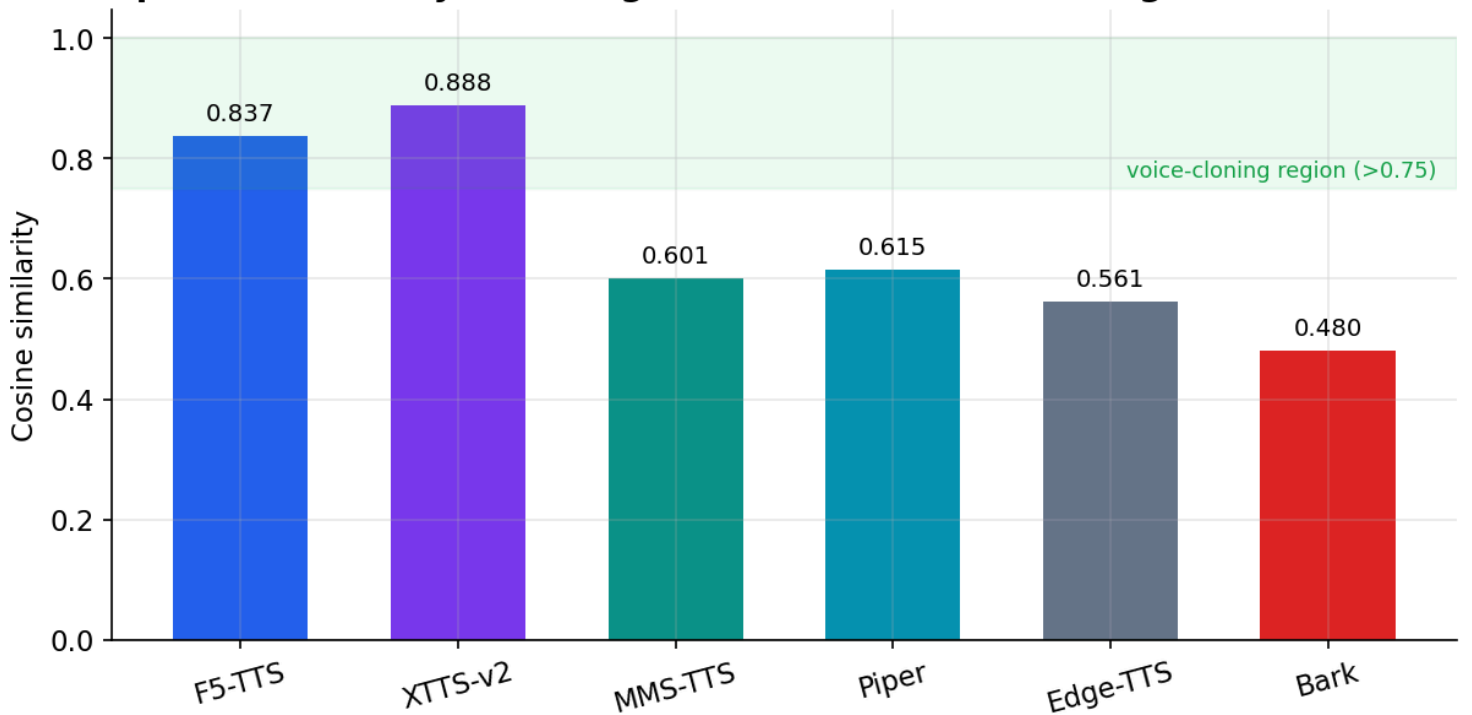


Figure 2. Speaker similarity (SECS). Only F5-TTS (0.837) and XTTS-v2 (0.888) enter the voice-cloning region (>0.75). Fixed-voice engines score ≈ 0.56 – 0.62 , reflecting a default voice rather than the reference identity.

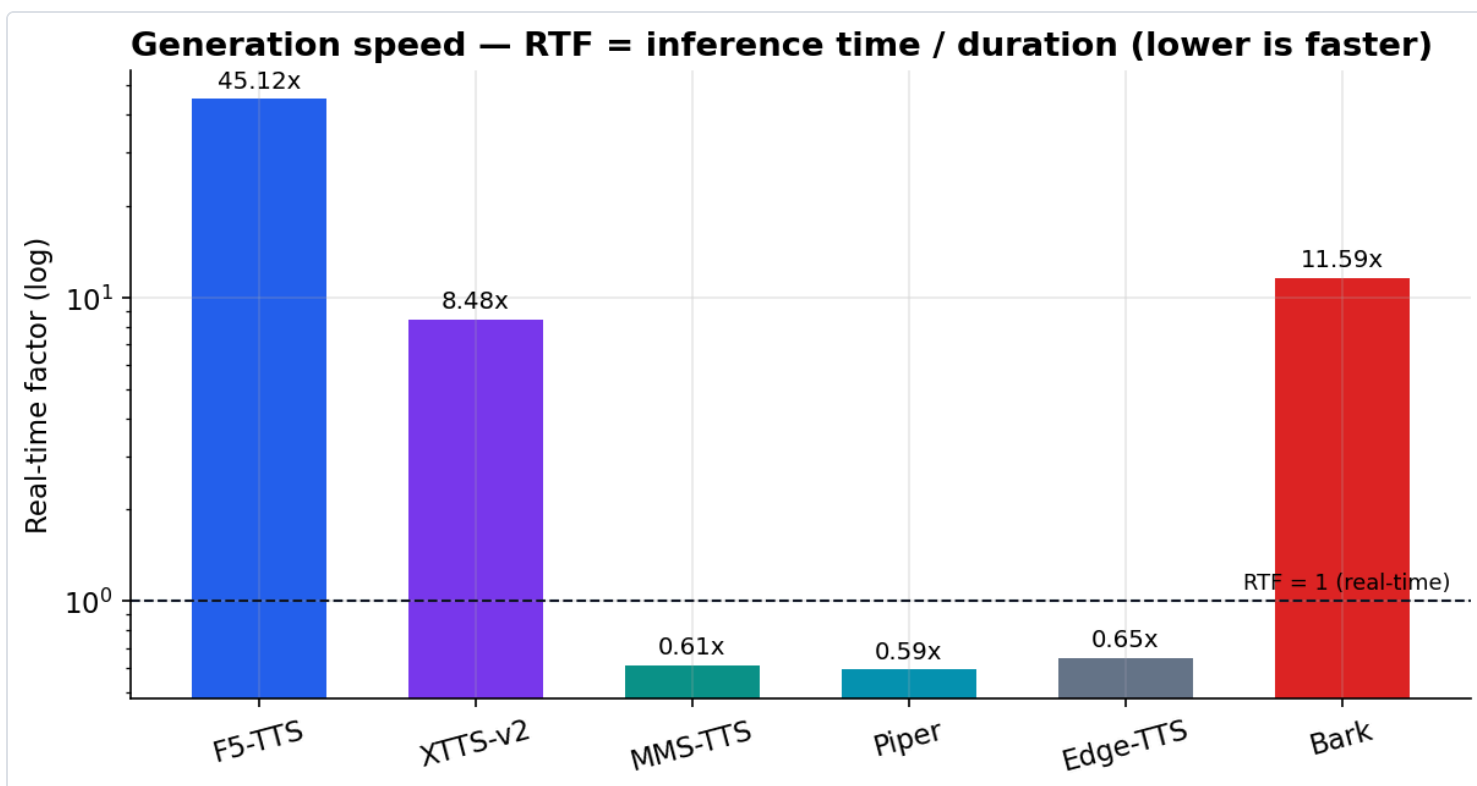
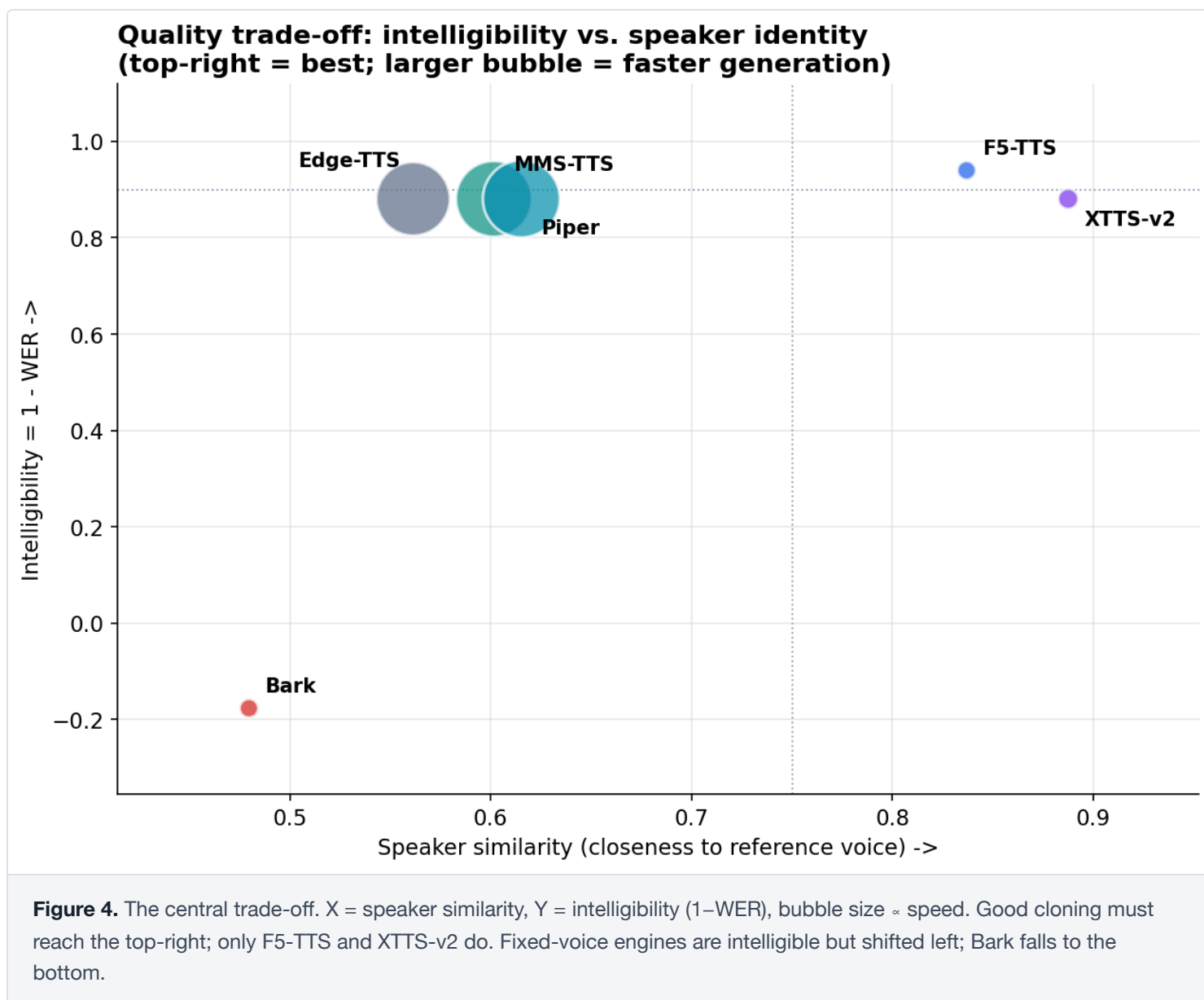


Figure 3. Generation speed (RTF, log scale). Piper, MMS, and Edge run faster than real time on CPU (RTF < 1); F5-TTS is slowest on CPU (RTF≈45), the cost of higher quality and multi-step flow matching — expected to drop sharply on GPU.



7 Discussion and Insights

F5-TTS wins on balance. It is the only engine that is simultaneously the most intelligible (WER 5.9%) and near the top on identity (SECS 0.837) — precisely the joint objective of voice cloning. **XTTS-v2 is a strong, faster alternative:** the highest SECS (0.888) and roughly 5× faster than F5-TTS on CPU, but with double the WER — the right choice when identity and speed outweigh accuracy.

Fixed-voice TTS is not cloning. MMS, Piper, and Edge are intelligible and very fast (RTF < 1) yet score SECS≈0.6, confirming they do not reproduce the reference speaker; they suit applications that only need clear Vietnamese. **CER stays below WER for every good engine** — F5-TTS errs chiefly on the sentence-initial word ("Trí"→"Chí", a *tr/ch* onset merge common in regional pronunciation), so CER (2.6%) is far below WER (5.9%): a dialectal pronunciation artifact, not an intelligibility failure. Finally, **the metrics catch real failure:** Bark is automatically flagged (WER > 100%, lowest SECS), demonstrating that the harness genuinely separates broken from working systems.

8 Limitations and Threats to Validity

- **Pilot-scale test** ($n = 1$ sentence/engine). This is an audit-scale study; it does not yet support confidence intervals or standard deviations. A ≥ 12 -sentence diverse set is prepared for the follow-up.
- **ASR is itself imperfect**. Whisper mis-transcribes, particularly tones, so WER/CER are upper bounds; read them comparatively.
- **SECS is relative**, not biometric verification.
- **Speed measured on CPU** and inclusive of first-load overhead for the first engine; absolute RTF will differ substantially on GPU.
- **Base model only**. Results characterize the zero-shot model; fine-tuning has not been verified end-to-end, so no fine-tuning improvement is claimed.

9 Conclusion and Future Work

We built and objectively benchmarked a Vietnamese zero-shot voice-cloning system. F5-TTS ViVoice is the right primary model: it leads on intelligibility and ranks in the top tier on speaker identity, the only engine to optimize both axes at once. Equally important, the project delivers a **reproducible objective-evaluation framework** that converts subjective impressions into verifiable measurements and that surfaced two subtle, high-impact defects during development.

Future work, in priority order: (1) scale the test set from one to ≥ 12 diverse sentences for mean $\pm$ standard-deviation reporting; (2) collect 30–60 minutes of real speaker audio and run fine-tuning end-to-end on GPU, measuring WER/CER/SECS before versus after adaptation; (3) add subjective Mean-Opinion-Score listening tests alongside the objective metrics; (4) optimize F5-TTS latency on GPU, including controlled NFE reduction; and (5) package the system (Docker / Hugging Face Spaces) for setup-free demonstration.

References

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Reproduce: `python scripts/evaluate.py --manifest evaluation/audit_existing_outputs.json` → `python scripts/plot_results.py --lang en`. All figures and Table 2 are generated from `outputs/evaluation/results.json`; no values are hand-edited. DSP391m
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